



# Stochastic Simultaneous Optimistic Optimization for Robotic Arm Simulator

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# Outline

Stochastic Simultaneous Optimistic Optimization (StoSOO)

StoSOO on the Synthetic Data

StoSOO for the TUDA Robotic Arm Simulator

# SOO Recap

- ▶ **SOO**: R. Munos. *Optimistic optimization of deterministic functions without the knowledge of its smoothness*. NIPS 2011.
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- ▶ **simultaneously** expands all leaves that can lead to potentially optimal solution
- ▶ **deterministic** – assumes that the evaluations of the unknown function are deterministic
- ▶ What if the function evaluations are *stochastic*?

# Comparison

|                    | Deterministic | Stochastic  |
|--------------------|---------------|-------------|
| known smoothness   | DOO           | StOO or HOO |
| unknown smoothness | Soo           |             |

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| unknown smoothness | Soo           | <b><i>StoSoo</i></b><br>this talk |

# Pseudocode of SOO

**while** True **do**

Set  $v_{\max} = -\infty$ .

**for**  $h = 0$  to  $\min(\text{depth}(\mathcal{T}_t), h_{\max}(t))$  **do**

Among all leaves  $(h, j) \in \mathcal{L}_t$  of depth  $h$ , select

$(h, i) \in \arg \max_{(h, j) \in \mathcal{L}_t} f(x_{h, j})$

**if**  $f(x_{h, i}) \geq v_{\max}$  **then**

Expand this node: add to  $\mathcal{T}_t$  the  $K$  children

$\{(h+1, i_1), \dots, (h+1, i_K)\}$  and evaluate the function at the corresponding centers  $\{x_{h+1, i_1}, \dots, x_{h+1, i_K}\}$

Set  $v_{\max} = f(x_{h, i})$ , Set  $t = t + 1$

**if**  $t = n$  **then Return**  $x(n) \stackrel{\text{def}}{=} \arg \max_{(h, i) \in \mathcal{T}_n} f(x_{h, i})$

**end if**

**end for**

**end while**



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$$b_{h,j}(t) \stackrel{\text{def}}{=} \hat{\mu}_{h,j}(t) + \sqrt{\frac{\log(n^2/\delta)}{2T_{h,j}(t)}}$$

where  $T_{h,j}(t)$  is the number of times  $(h,j)$  has been selected up to time  $t$ , and  $\hat{\mu}_{h,j}(t)$  is the empirical average of rewards

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- ▶ under some assumptions, for  $k = n/\log^3 n$ ,

$$\mathbb{E}[r_n] = O\left(\frac{\log^2 n}{\sqrt{n}}\right)$$

# Pseudocode of StoSOO

```

while  $t \leq n - \min(\text{depth}(\mathcal{T}_t), h_{\max}(t))$  do
  Set  $v_{\max} = 0$ .
  for  $h = 0$  to  $\min(\text{depth}(\mathcal{T}_t), h_{\max}(t))$  do
    Among all leaves  $(h, j) \in \mathcal{L}_t$  of depth  $h$ , select
     $(h, i) \in \arg \max_{(h, j) \in \mathcal{L}_t} b_{h, j}(t)$ 
    if  $b_{h, i}(t) \geq v_{\max}$  then
      Sample state  $x_t = x_{h, i}$  and collect reward  $r_t$ 
      if  $T_{h, i}(t) \geq k$  then
        Expand this node: add to  $\mathcal{T}_t$  the  $K$  children of  $(h, i)$ 
        Set  $v_{\max} = b_{h, i}(t)$ .
        Set  $t \leftarrow t + 1$ .
      end if
    end if
  end for
end while.
Return the state corresponding to the deepest expanded node:

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$$x(n) = \arg \max_{x_{h, j}: (h, j) \in \mathcal{T}_n \setminus \mathcal{L}_n} h.$$

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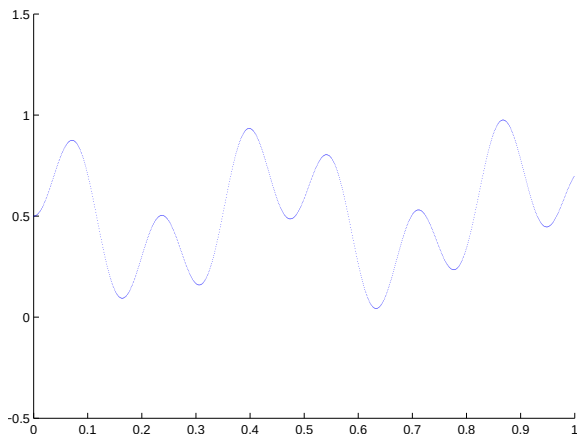


# Sine Function

$$f(x) = \frac{\sin(13x) \cdot \sin(27x)}{2} + 0.5$$

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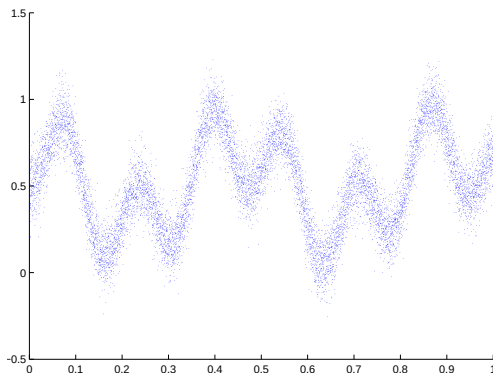


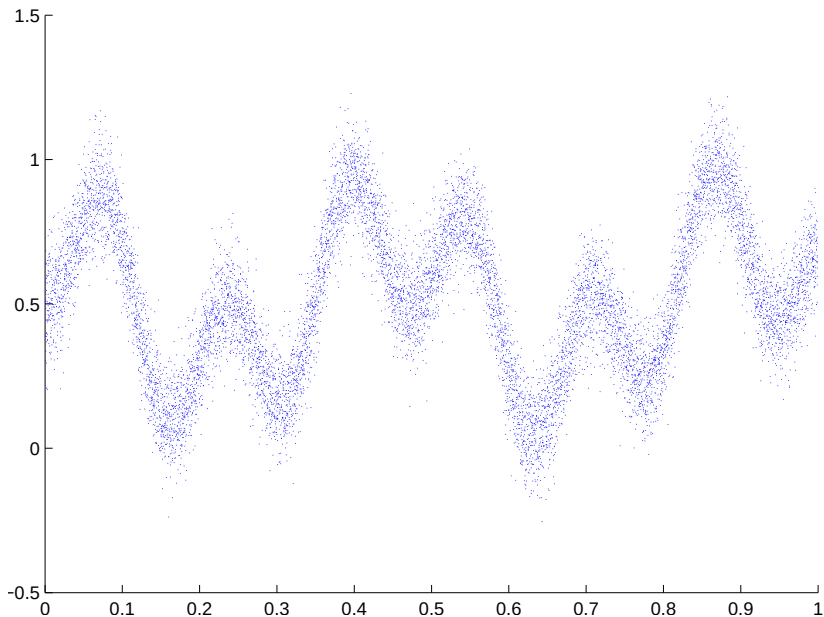
# Noised Sine Function

$$f(x) = \frac{\sin(13x) \cdot \sin(27x)}{2} + \mathcal{N}(0.5, 0.1)$$

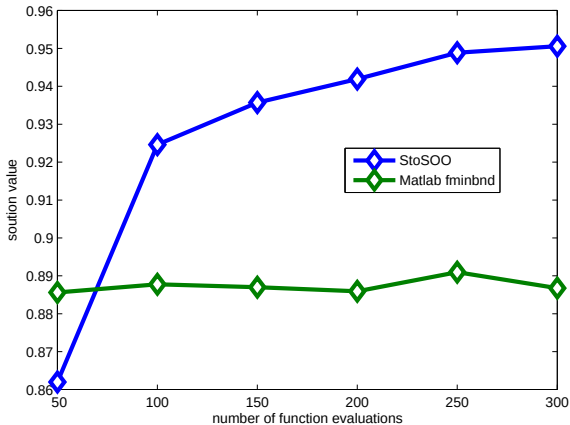
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# Noised Sine Function: Results



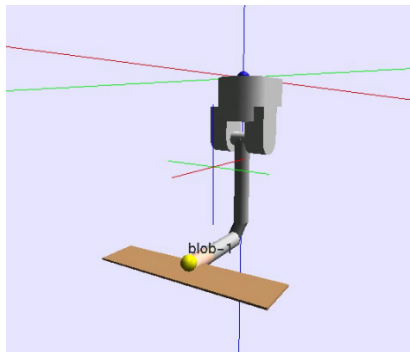
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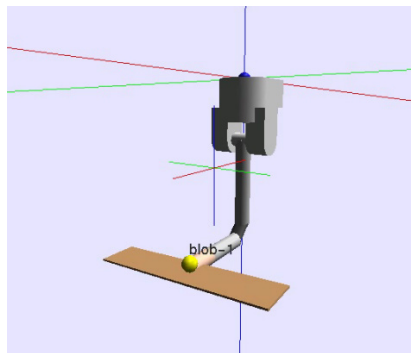
StoSOO for the TUDA Robotic Arm Simulator

# Ball on the Beam Policy Optimization



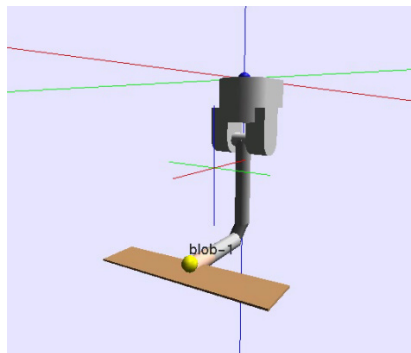


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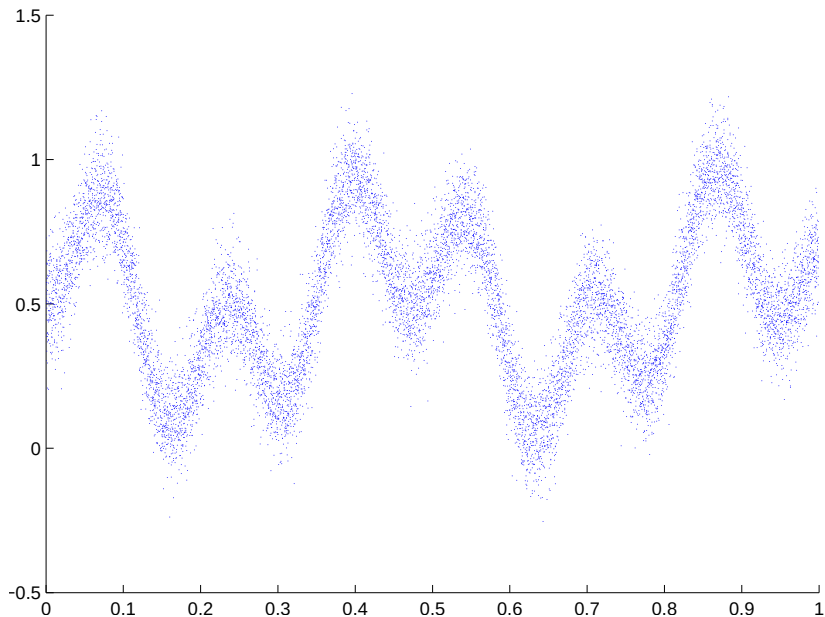


- **Task:** optimize parameters  $(c_1, c_2, c_3)$  of the parametric policy  $\theta(x, \dot{x}, \ddot{x}) = c_1x + c_2\dot{x} + c_3\ddot{x}$ , where  $x$  is the distance between the center of the plate ( $m$ ) and the ball

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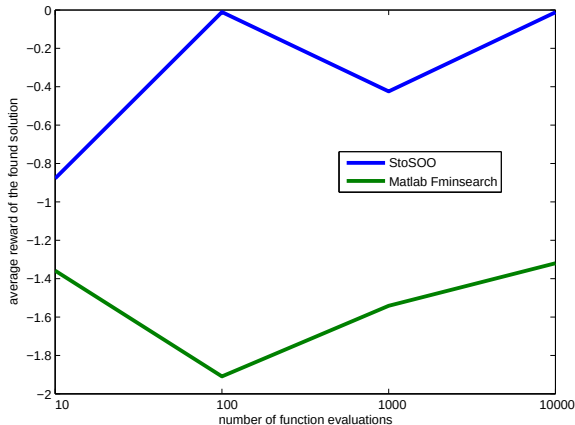
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- **Reward**  $= \sum_t (x_t - m)^2$



# Subcomponents for the Compositional Approach

- ▶ stochastic function **optimizer** with unknown smoothness
- ▶ TUDA robotic arm **simulator** (input: *policy*, output: *reward*)
- ▶ **representation** module that provides features or the parameterization (this is given for now)

# Ball on the Beam Policy Optimization



# Demonstration of the Policy

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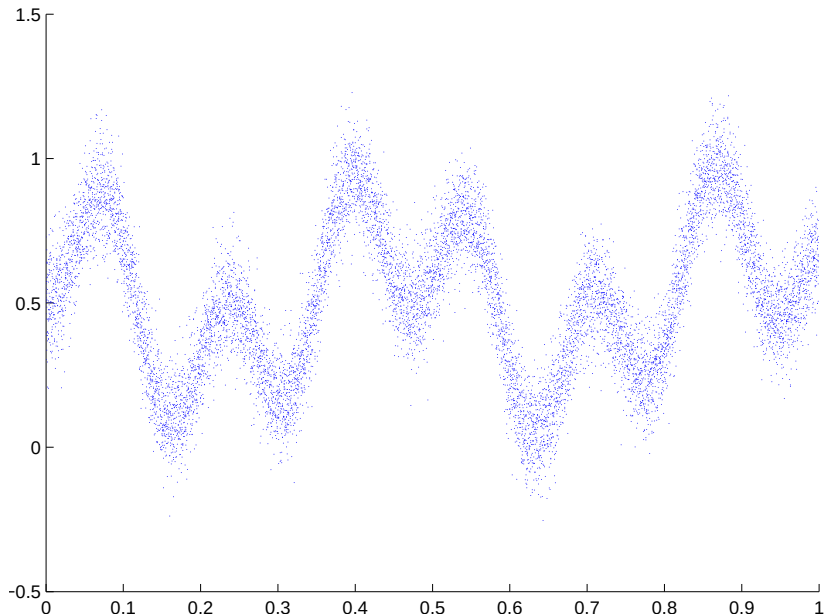
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- ▶ StoSOO after 10000 iterations:  
 $c_1 = 0.834, c_2 = 0.944, c_3 = 0.056$
- ▶  $E[\text{Reward}(c_1 = 0.834, c_2 = 0.944, c_3 = 0.056)] = -0.011$   
(*video*)

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# Thank you!

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